Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)	The energy of the photon becomes (1) PE and KE of electron (accept freeing electron for PE) (1) **Alternative:** Photon energy = KE + workfunction (1) Photon energy = KE + energy to release electron (2)	2			2		
		(ii)	Current / no. of electrons per sec / charge per sec etc. Accept power			1	1		
	(b)		Substituting correctly for r in equation 3 e.g. $p^2 = \frac{m_e e^2}{4\pi\varepsilon_0 \frac{nh}{2\pi p}}$(1) Correct algebra leading to $p = \frac{m_e e^2}{2\varepsilon_0 nh}$(1) (Minimum algebra: $p^{\cancel{2}} = \frac{m_e e^2}{\cancel{4\pi}\varepsilon_0 \frac{nh}{\cancel{2\pi p}}}$ QED i.e. cancelling π and p)		2		2	2	
	(c)	(i)	Minimum of 3 values substituted correctly (1) Answer = 13.55 [eV] or final substitution seen (1)	1	1		2	2	
		(ii)	Explain n = 1 e.g. leaving n = 1 or substituting n = 1 (1) Explain $n = \infty$ or PE at ∞ (1)		2		2		
	(d)		Valid method adopted e.g. using n = 3 and n = 2 to calculate the wavelength **OR** using 656 nm to confirm energy gap between n = 2 and 3 (1) $E = \frac{hc}{\lambda} = 3.03 \text{ x } 10^{-19}$ [J] **OR** $13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right)$ **OR** 1.89 (1) Final confirmation seen e.g. $3.03 \text{ x } 10^{-19}$J = $13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right) \times 1.6 \times 10^{-19}$ **OR** $\frac{hc}{13.6 \times \left(\frac{1}{4} - \frac{1}{9}\right) \times 1.6 \times 10^{-19}}$= 656 n[m] (1)			3	3	2	

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	(e)		Easier to escape **or** more electrons escaping **or** tunnelling linked to electrons escaping (1) Light at lower V **or** <u>measured</u> V is lower **or** actual barrier is higher (1)		2		2		
	(f)	(i)	Electrons have greater [internal] energy i.e. link between temperature and energy (1) More electrons (1)		2		2		
		(ii)	Use of $\frac{3}{2}kT$ can be implied (1) Conversion to eV (1) Answer = 0.038 [eV] (1) Diode switches on 0.038 V too early or 0.038 V is an uncertainty - any good sensible comment regarding this equivalent pd (1)			4	4	2	
			Question 8 total	**3**	**9**	**8**	**20**	**8**	**0**